

MU123

TMA 02

2025J

Covers Units 3, 4 and 5

Cut-off date 16 December 2025

Submission instructions

You should submit this TMA electronically as a PDF file by using the University's online TMA/EMA service. Before starting work on it, please read the guidance for preparing and submitting TMAs, available from the 'Assessment' tab of the module website.

Reviewing your tutor's comments on your previous TMA will help you as you work on this one.

Special instructions

Remember that you need to explain your reasoning and communicate your ideas clearly, as described in Subsection 5.3 of Unit 1. This includes:

- explaining your mathematics in the context of the question
- the correct use of notation and units
- appropriate rounding.

Your score out of 5 for good mathematical communication (GMC) will be recorded against Question 6. You do not have to submit any work for Question 6.

PLAGIARISM WARNING – the use of assessment help services and websites

The **work that you submit for any assessment on any module should be your own**. Submitting work produced by or with another person, or a web service or an automated system, **as if it is your own** is cheating. It is **strictly forbidden** by the University.

You should not:

- provide any assessment question to a website, online service, social media platform or any individual or organisation, as this is an infringement of copyright
- request answers or solutions to an assessment question on any website, via an online service or social media platform, or from any individual or organisation
- use an automated system (other than one prescribed by the module) to obtain answers or solutions to an assessment question and submit the output as your own work
- discuss examination questions with any other person, including your tutor.

The University actively monitors websites, online services and social media platforms for answers and solutions to assessment questions, and for assessment questions posted by students. Work submitted by students for assessment is also monitored for plagiarism.

A student who is found to have posted a question or answer to a website, online service or social media platform and/or to have used any resulting, or otherwise obtained, output as if it is their own work has committed a disciplinary offence under our [Code of Practice for Student Discipline](#). **This means the academic reputation and integrity of the University has been undermined.**

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- using text obtained from assignment writing sites, organisations or private individuals
- obtaining work from other sources and submitting it as your own.

If it is found that you have used the services of a website, online service or social media platform, or that you have otherwise obtained the work you submit from another person, this is considered serious academic misconduct and you will be referred to the Central Disciplinary Committee for investigation.

Question 1 – 5 marks

This question is designed to help you and your tutor understand your current approach to studying MU123, and how best to plan ahead. Please write a sentence or two for each part.

- (a) What have you enjoyed most about your MU123 experience so far? [1]
- (b) What has been your most challenging experience on MU123 so far? [1]
- (c) What is the most important thing you have learnt from your tutor's feedback from TMA01? [1]
- (d) How has this changed the way you approached this assignment? [2]

If, for any reason, you don't have access to feedback from TMA01, explain why, and say how you have tried to implement the advice in pages 60 - 64, Book A, about Good Mathematical Communication in this TMA.

Question 2 – 25 marks

This question is based on your work on MU123 up to and including Unit 3.

- (a) (i) Use a factor tree or similar method to write 8316 as a product of prime factors. Display your factor tree in your answer. If you choose to use an alternative method, then clearly show your working: just writing down the answer is not sufficient. [4]
- (ii) Calculate
$$\frac{5}{6} - \frac{2}{7} + \frac{13}{21},$$
leaving your answer as a fraction in its simplest form, showing your working. [4]
- (iii) Simplify the surd
$$\frac{5\sqrt{8}}{\sqrt{40}} + \sqrt{80}$$
by writing it as a surd in its simplest form, showing your working. [3]
- (b) 540 students are required to take a test. Of these, 495 actually take the test, and 45 are absent that day.
 - (i) Write down the ratio of the number of students who take the test to the number of students who are absent. Simplify this ratio as far as possible. [3]
 - (ii) Of the 495 students who take the test, the numbers of students obtaining grades A, B, C and D are in the ratio 8 : 12 : 9 : 4, respectively. Calculate the number of students obtaining each grade. Show how you could check that your answers are plausible. [4]

- (c) Jannik Sinner played in tennis matches for a total of 17 hours and 46 minutes to win the Australian Open men's singles title in 2025. The winner's prize money was \$3 500 000 (Australian dollars).
- (i) Calculate how long Jannik played (in minutes) for each Australian dollar that he won in this tournament.
Give your answer to three significant figures in both ordinary notation and scientific notation. [4]
- (ii) Calculate how much Jannik won (in Australian dollars) for each hour that he played in this tournament.
Give your answer to three significant figures in both ordinary notation and scientific notation. [3]

Question 3 – 25 marks

This question is based on your work on MU123 up to and including Unit 4.

An IT company develops software for the Agricultural sector. It tests its code using an industry standard software package. Aiming to expand its business and be more competitive in the sector, the company has decided to pilot an in-house software package to test its code. A company researcher wishes to compare the times taken for each package to complete tests. Both software packages were run on 20 occasions testing the same code. The researcher records the times taken on each occasion, and these are shown in Table 1.

Table 1 Comparison of test times in minutes for the industry standard and in-house software testing packages

Test number	Industry standard software package	In-house software package
1	438	430
2	442	450
3	450	412
4	467	420
5	448	415
6	498	418
7	410	450
8	453	438
9	654	410
10	458	435
11	420	440
12	438	418
13	472	432
14	480	445
15	450	425
16	480	445
17	405	385
18	440	420
19	475	450
20	472	442

- (a) (i) Which of the three types of investigation discussed on pages 210–214 of Unit 4 is this? Explain your answer briefly. [2]
- (ii) Are these primary or secondary data, from the researcher’s point of view? Explain your answer briefly. [2]
- (b) List any three features that you should be looking for when initially scanning datasets by eye. [3]
- (c) Copy and complete the following table, using the datasets given above, to work out the missing values. [4]

	Number of minutes to complete the test	
	Industry standard	In-house
Minimum (Min)		
Lower quartile (Q1)	439	418
Median	451.5	431
Upper quartile (Q3)	473.5	443.5
Maximum (Max)		
Mean	462.5	429
Standard deviation (SD)	49.76	16.51
Interquartile range (IQR)		
Range		
Size of dataset (n)	20	20

- (d) (i) Identify the two measures of location from the table in part (c). By referring to both of these measures, explain which of the two datasets has the higher location. [4]
- (ii) Identify the three measures of spread from the table in part (c). By referring to each of these three measures, explain which of the two datasets has the wider spread. [4]
- (e) (i) The researcher concludes that the in-house software package runs quicker than the industry standard software package. Is this a reasonable conclusion? Explain your answer briefly. [2]
- (ii) Which stage of the statistical investigation is used in part (e)(i)? Briefly justify your answer. [2]
- (f) The researcher notices that the industry standard software package entry for the ninth test was a typing error. The correct value should have been 454. The revised mean and median for the industry standard software package data with the correct value for the ninth test are given in the following table.

	Industry standard software package	
	With typing error	With correct value
Mean	462.5	452.5
Median	451.5	451.5

What is the effect on the mean and on the median of including the typing error instead of the correct value? Explain why this happens. [2]

Question 4 – 15 marks

This question is based on your work on MU123 up to and including Unit 5.

In this question, you are asked to comment on a student's incorrect attempt at answering the question detailed below.

The question

- (i) Simplify the following expression as far as possible, showing all your working:

$$12x - 17 - 7x + 72.$$

- (ii) Solve the following equation, showing all your working:

$$\frac{5}{12}(t - 4) = 10.$$

The student's incorrect attempt:

(i)

Given	$12x - 17 - 7x + 72$
Group like terms	$= 12x - 7x - 17 + 72$
Collect like terms	$= 5x + 55$
Subtract 55	$- 55 = 5x$
Divide by 5	$- 11 = x$

The solution is $x = -11$

(ii)

Given	$\frac{5}{12}(t - 4) = 10$
Clear the fraction	$5(t - 48) = 120$
Multiply by 5	$5t - 240 = 120$
Add 240	$5t = 360$
Divide by 5	$t = 72$

The solution is $t = 72$

- (a) The student has made a mistake in both parts of their attempt.

Identify each mistake and explain, as if directly to the student, why, for each mistake, their reasoning is incorrect.

[4]

- (b) Write out your own solution to each part of the question, showing your working.

[6]

- (c) The student decides to check if the solution they got to part (ii) of the question was correct using the following approach:

Check

$$\text{Substitute } 72 \text{ for } t \quad \frac{5}{12}(72 - 4) = 10$$

$$\text{Work out the brackets} \quad \frac{5}{12}(68) = 10$$

$$\text{Giving} \quad \frac{340}{12} = 10$$

This is incorrect so the solution $t = 72$ is incorrect

- (1) Explain why *the approach to checking* that they have chosen is incorrect. [2]
- (2) Show how the student should have performed the check and explain your approach to checking. [3]

Question 5 – 25 marks

This question is based on your work on MU123 up to and including Unit 5.

- (a) Simplify each of the following expressions as far as possible; multiply out any brackets, expand any algebraic fractions, and collect like terms together. Show your working.
- (i) $7(9 - 3t)$ [2]
 - (ii) $31 + m(15 - 9m) - 15m$ [3]
 - (iii) $36t - 9(11s - 7t)$ [3]
 - (iv) $7p(5 + 12p^2) - 4(6p - 8 + 3p^3)$ [4]
 - (v) $\frac{7x^2 + 42x - 63}{7x}$ [4]
- (b) Solve the following equations. Show your working and show that you have checked that your solutions are correct.
- (i) $21a - 18 = 36 + 12a$ [4]
 - (ii) $\frac{y}{5} - 4 = 3(y - 6)$ [5]

Question 6 – 5 marks

A score out of 5 marks for good mathematical communication over the entire TMA will be recorded under Question 6. [5]